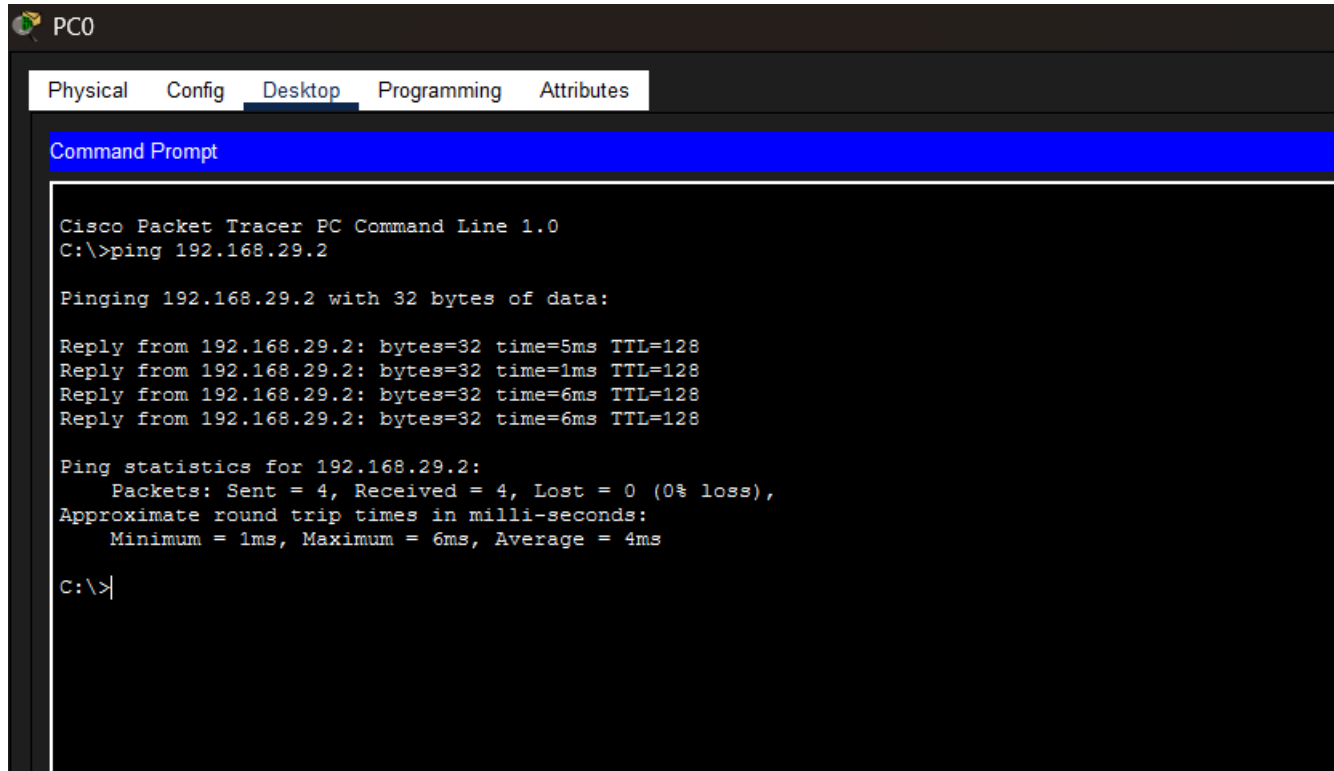


Q-1. Use the ping command to test connectivity between two PCs on different subnets in your Packet Tracer network and document the output for both successful and failed attempts.



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is selected, and the 'Command Prompt' application is open. The command prompt displays the following output:

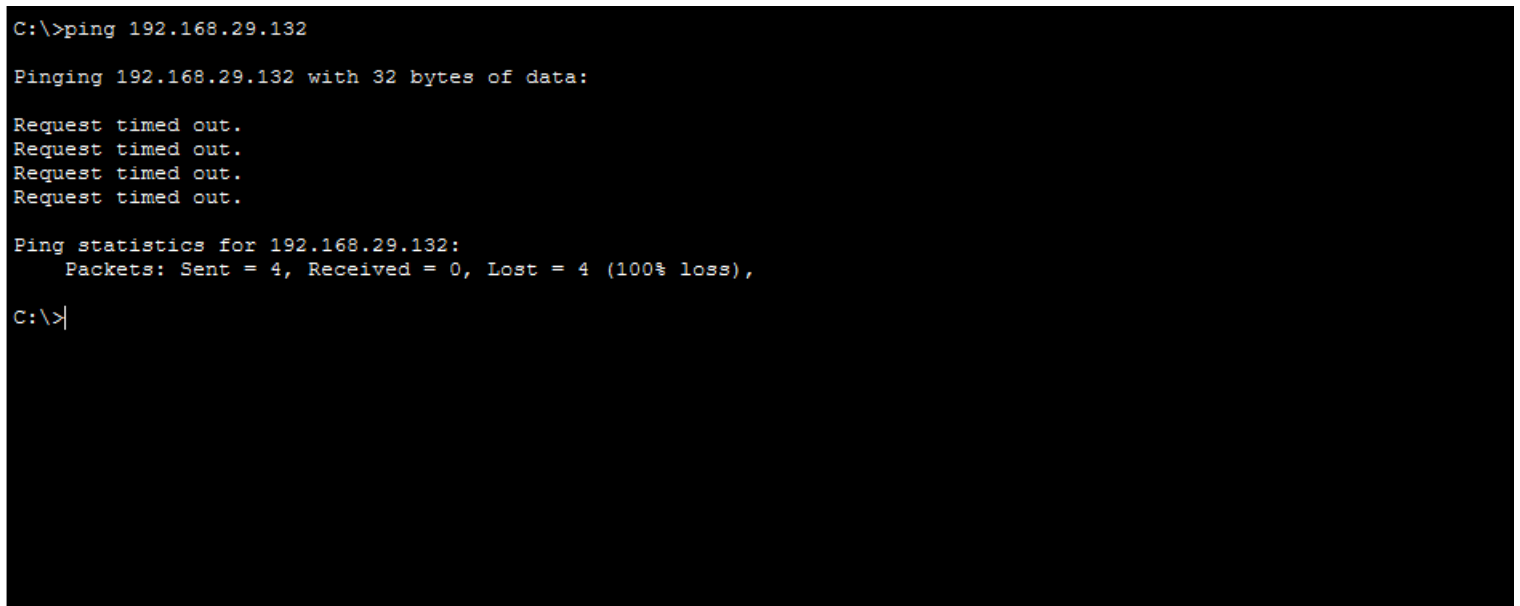
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.2

Pinging 192.168.29.2 with 32 bytes of data:

Reply from 192.168.29.2: bytes=32 time=5ms TTL=128
Reply from 192.168.29.2: bytes=32 time=1ms TTL=128
Reply from 192.168.29.2: bytes=32 time=6ms TTL=128
Reply from 192.168.29.2: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.29.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 4ms

C:\>
```



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is selected, and the 'Command Prompt' application is open. The command prompt displays the following output:

```
C:\>ping 192.168.29.132

Pinging 192.168.29.132 with 32 bytes of data:

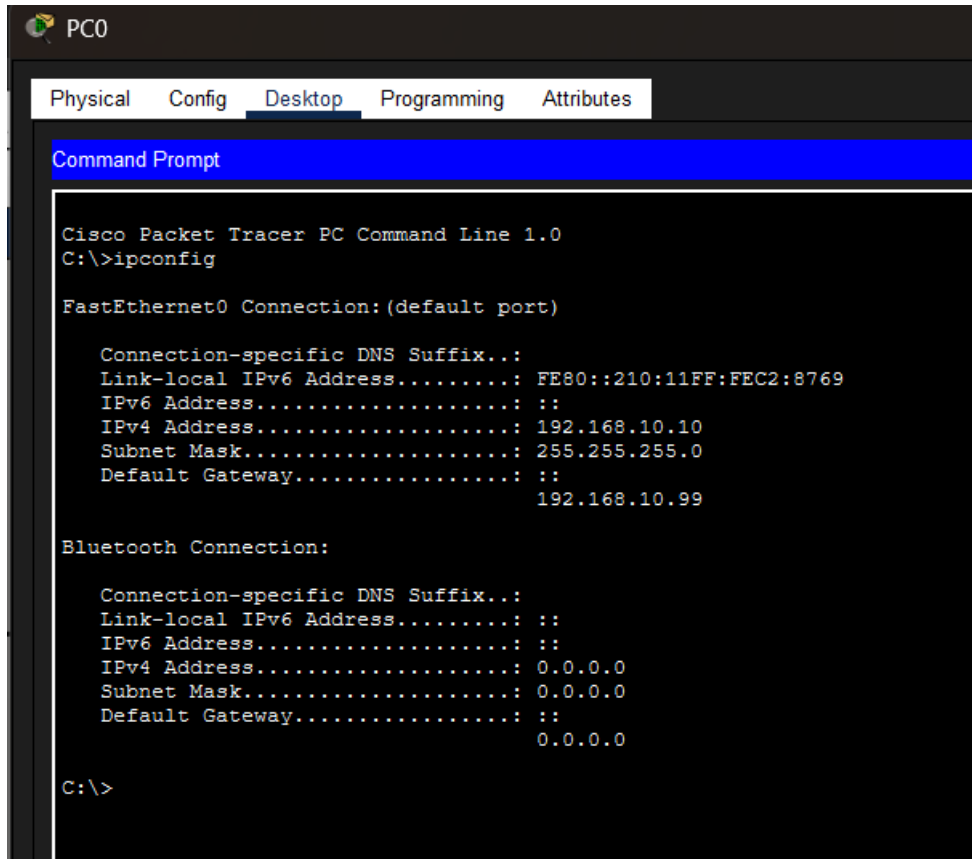
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.29.132:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

a Layer-2 switch forwards frames within the same broadcast domain. Communication between different IP subnets requires a **Layer-3 device**, such as a router or multilayer switch.

Q-2. Identify and fix an incorrect default gateway configuration on one device in your network so that it can access an external web server simulation; submit the steps you took and the command outputs before and after the fix.



The screenshot shows the Cisco Packet Tracer interface for PC0. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the output of the 'ipconfig' command, showing the configuration for the FastEthernet0 interface. The IPv4 address is 192.168.10.10, the subnet mask is 255.255.255.0, and the default gateway is 192.168.10.99. The Bluetooth connection is also shown with its configuration details.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

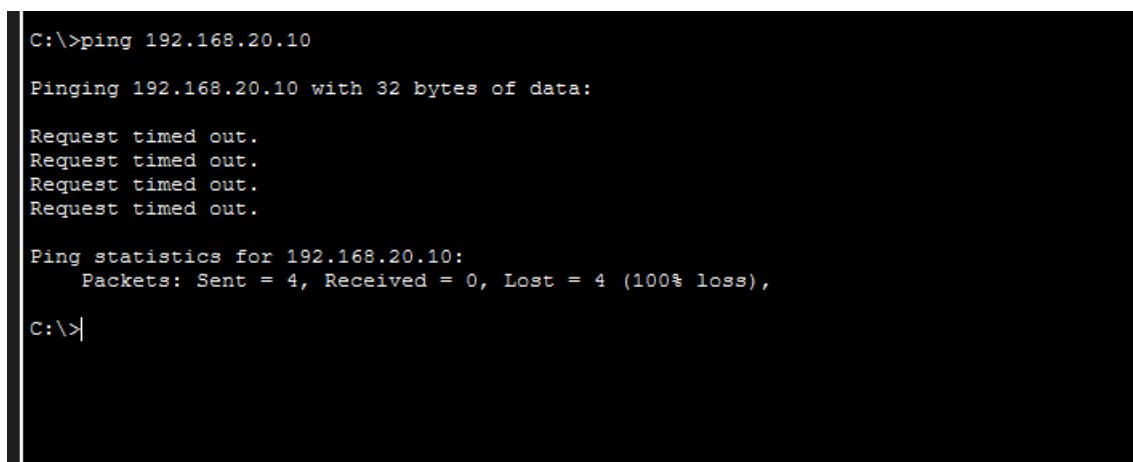
FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::210:11FF:FEC2:8769
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.10
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.10.99

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```



The screenshot shows a Command Prompt window with the command 'ping 192.168.20.10' entered. The output shows four 'Request timed out' messages, indicating that the ping failed. The ping statistics show that 4 packets were sent, 0 were received, and 4 were lost (100% loss).

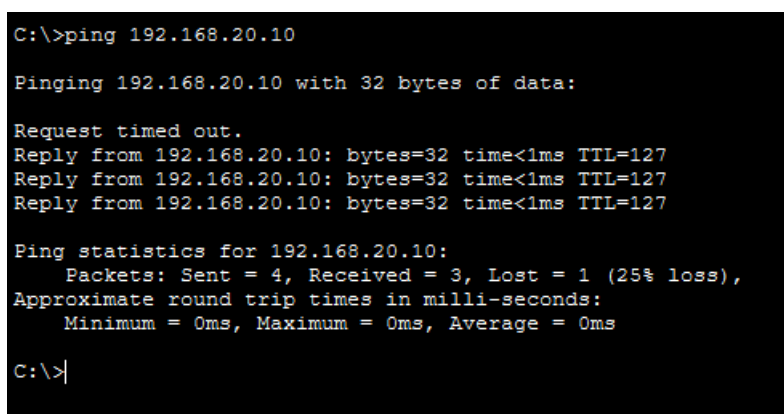
```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



The screenshot shows a Command Prompt window with the command 'ping 192.168.20.10' entered. The output shows three successful replies from 192.168.20.10, indicating that the ping succeeded. The ping statistics show that 4 packets were sent, 3 were received, and 1 was lost (25% loss). The approximate round trip times in milliseconds are also shown.

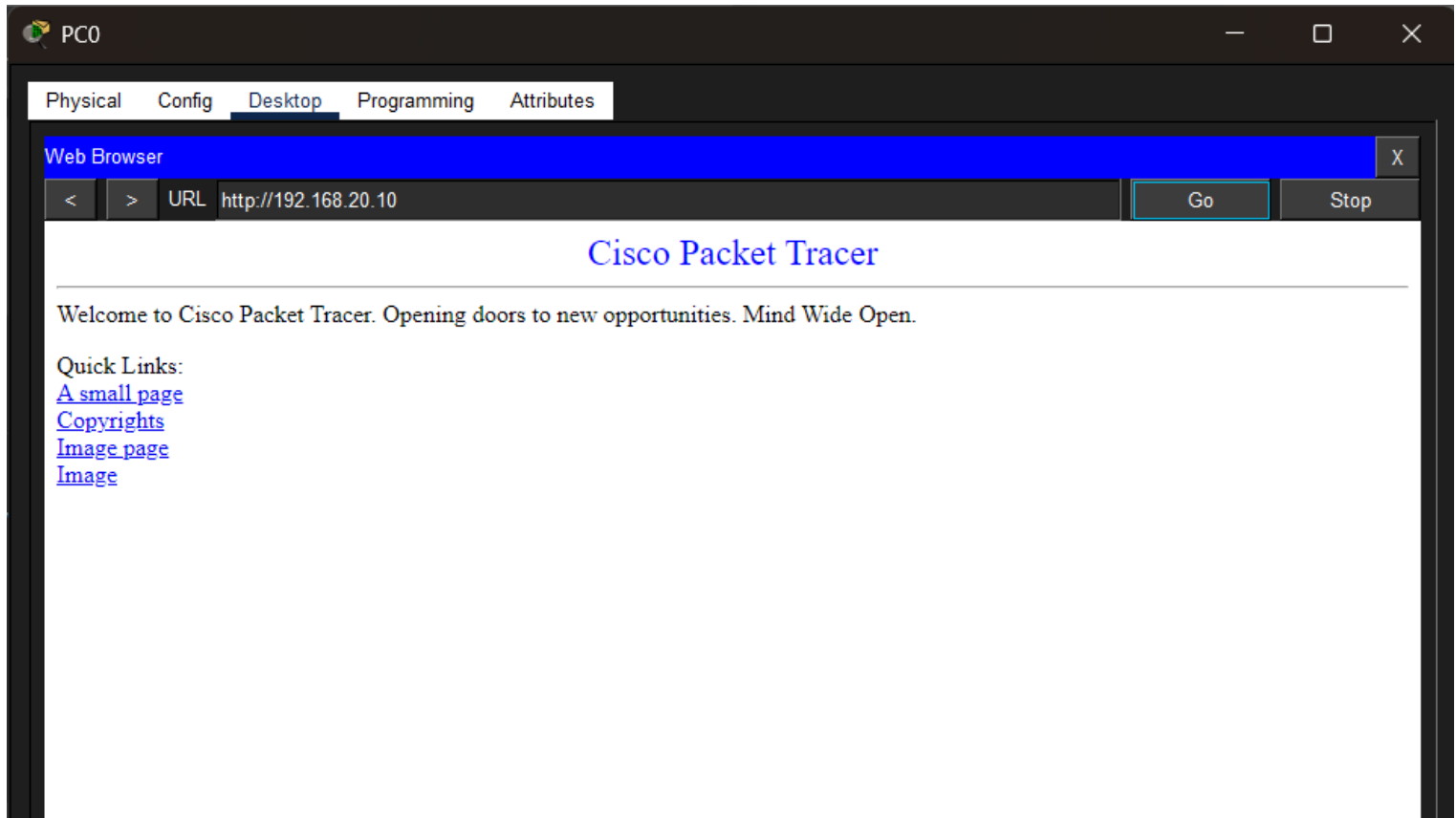
```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

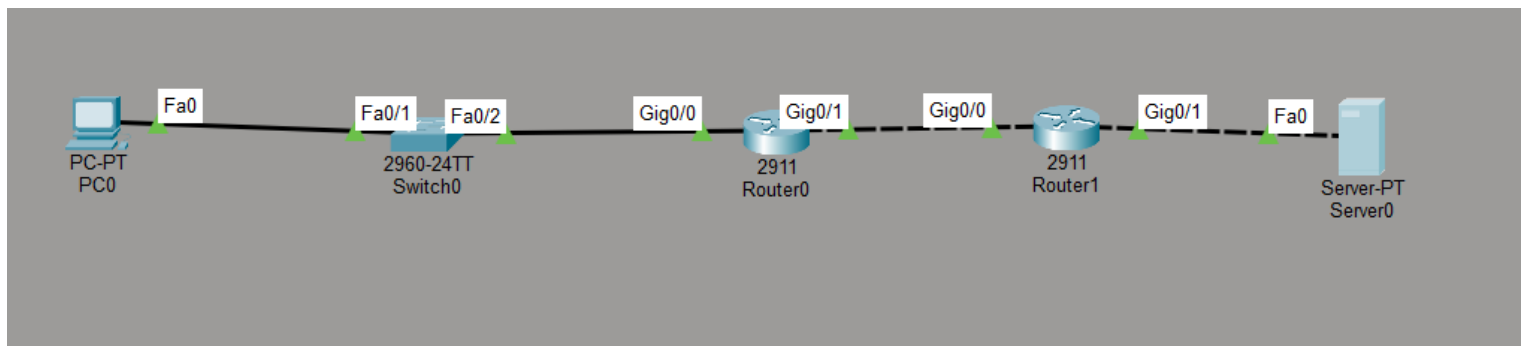
Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



The default gateway was incorrectly configured as 192.168.10.254. As a result, the PC could not reach the web server 192.168.20.2, resulting in 100% packet loss.

Q-3 A user reports they cannot access Flipkart.com from their simulated network. Trace the path using traceroute (or tracert) and identify where the connection fails; explain how you diagnosed and resolved the issue.



IP address

Device	Interface	IP Address
PC0	Fa0	192.168.10.10
Router0	G0/0	192.168.10.1

Device	Interface	IP Address
Router0	G0/1	10.0.0.1
Router1	G0/0	10.0.0.2
Router1	G0/1	192.168.20.1
Server0	Fa0	192.168.20.10

Test Before Fix

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert 192.168.20.10

Tracing route to 192.168.20.10 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.10.1
  2  *          *          *          Request timed out.
  3  *          *          *          Request timed out.
  4  *          *          *          Request timed out.
  5  *          *          *          Request timed out.
  6  *          *          *          Request timed out.
  7  *          *          *          Request timed out.
  8  *          *          *          Request timed out.
  9  *          *          *          Request timed out.
 10 *          *          *          Request timed out.
 11 *          *          *          Request timed out.
 12 *          *          *          Request timed out.
 13 *          *          *          Request timed out.
 14 *          0 ms      0 ms      192.168.20.10

Trace complete.
```

Diagnosis

I checked Router1's routing table using:

show ip route

Router1 did not have a route to the PC0 network:

192.168.10.0/24

Therefore, the server's reply could not return to PC0.

Fix

On **Router1**, I added the missing route:

enable

configure terminal

```
ip route 192.168.10.0 255.255.255.0 10.0.0.1
end
```

Then verified it with:

```
show ip route
```

The routing table showed:

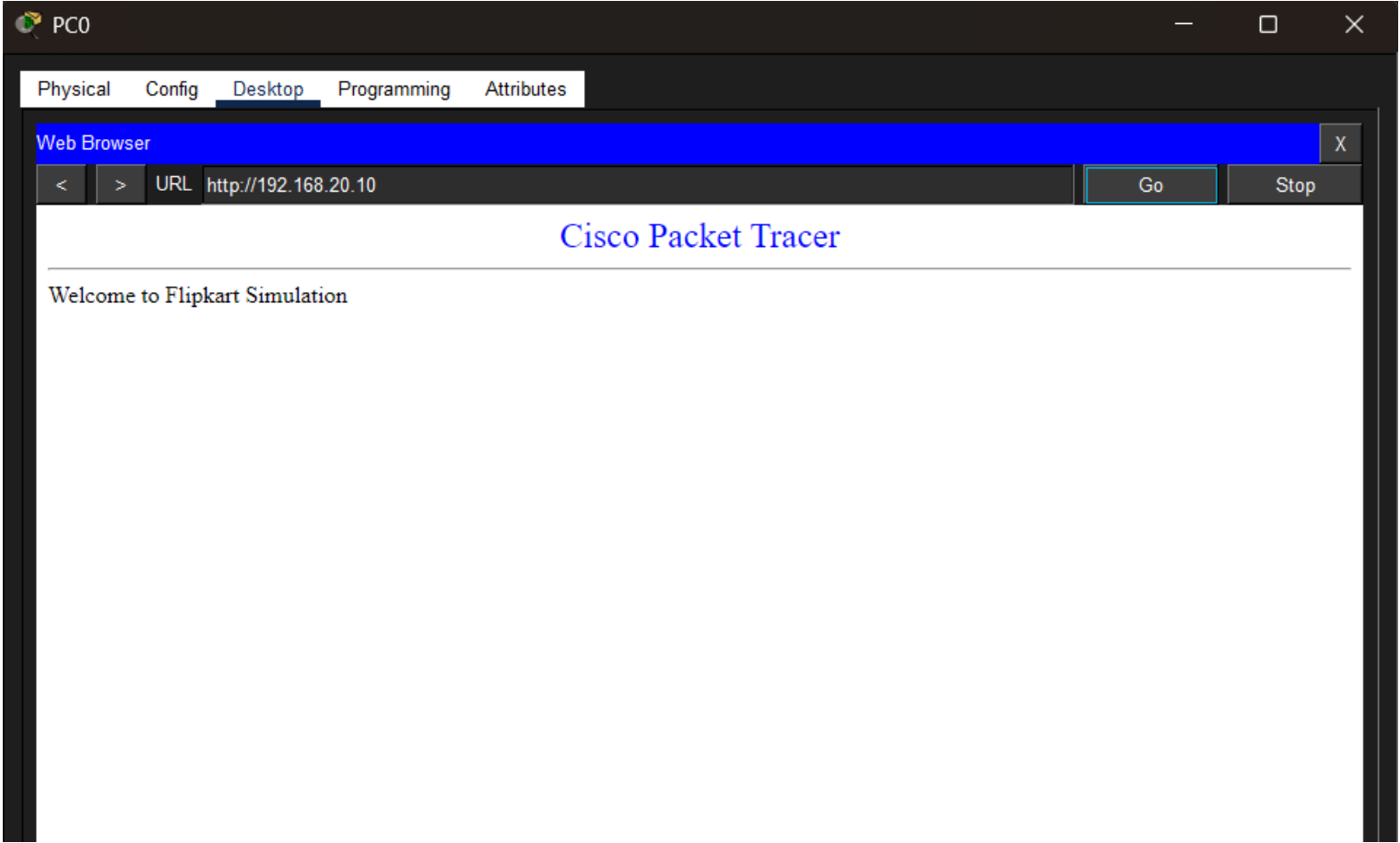
```
C:\>tracert 192.168.20.10

Tracing route to 192.168.20.10 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.10.1
  2  1 ms    1 ms    1 ms    10.0.0.2
  3  0 ms    1 ms    1 ms    192.168.20.10

Trace complete.

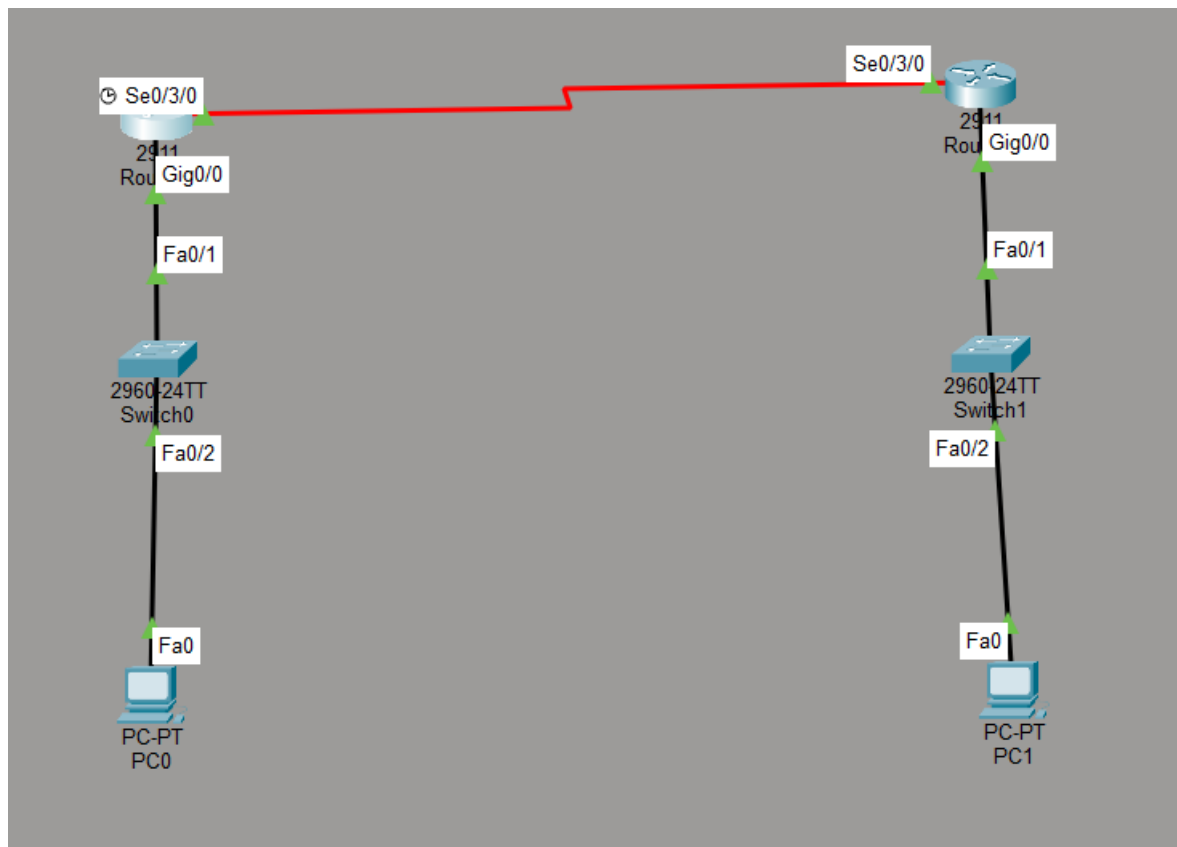
C:\>|
```



The connectivity problem was caused by a missing return route on Router1. I identified the failure using tracert, checked the routing table with show ip route, added the missing static route, and verified connectivity using ping and tracert.

**Q-4. Simulate a scenario where a router has an incorrect static route that breaks connectivity between two VLANs. Correct the route configuration and verify the solution using appropriate show commands.

Hint: Use 'show ip route' and 'show running-config' to check current routes and settings.**



Router 0

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.5
Router(config)#exit
Router#
```

Router 1

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial10/3/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial10/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial10/3/0, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Test connectivity

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Troubleshoot using show commands

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial10/3/0
L       10.0.0.1/32 is directly connected, Serial10/3/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
```

```

interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 duplex auto
 speed auto
 !
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
 !
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
 !
interface Serial0/3/0
 ip address 10.0.0.1 255.255.255.252
 clock rate 2000000
 !
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
 !
interface Vlan1
 no ip address
 shutdown
 !
ip classless
ip route 192.168.20.0 255.255.255.0 10.0.0.5
 !
ip flow-export version 9
 !
 !
 !

```

That's our incorrect static route Find.

Correct the static route

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip route 192.168.20.0 255.255.255.0 10.0.0.5
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

```

Verify the corrected

```

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

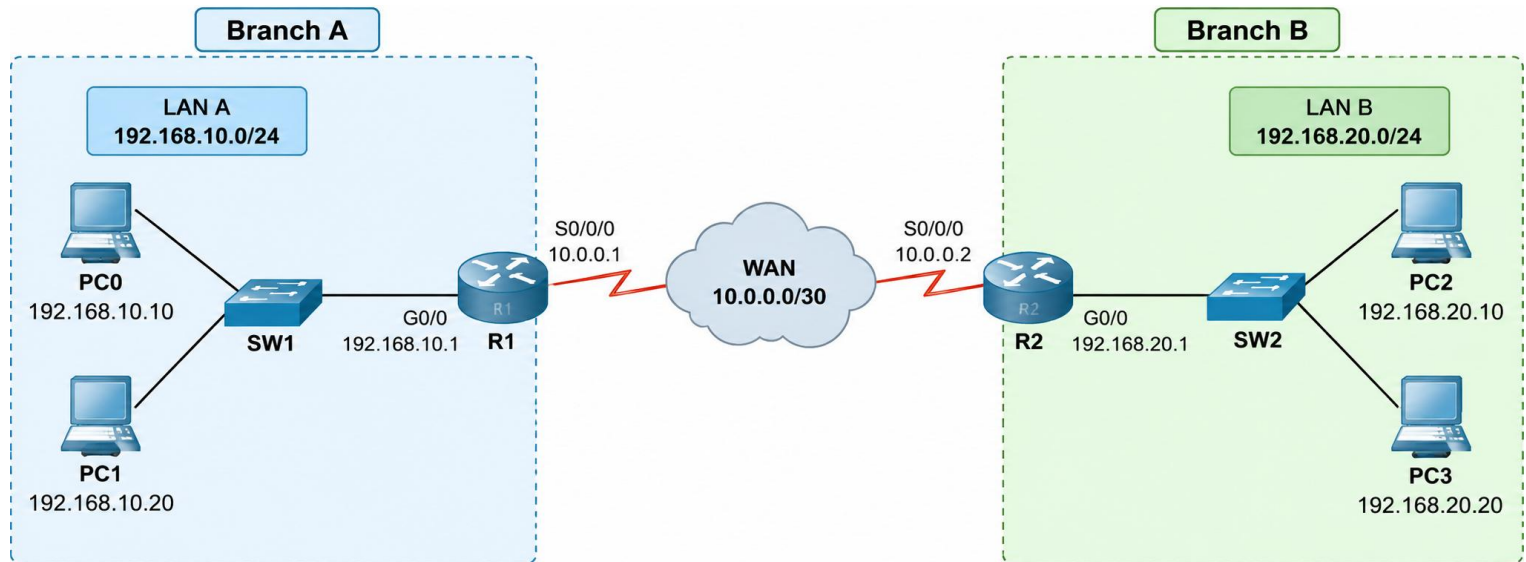
Request timed out.
Reply from 192.168.20.2: bytes=32 time=2ms TTL=126
Reply from 192.168.20.2: bytes=32 time=7ms TTL=126
Reply from 192.168.20.2: bytes=32 time=7ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 7ms, Average = 5ms

C:\>

```

Q-5.You receive a network diagram and a set of running-config outputs for a multi-router setup where WhatsApp voice calls are not working between two branches. Analyze the configs, identify the misconfiguration, and describe your troubleshooting steps and resolution.



Troubleshooting Steps

Step 1: Check Interface Status

show ip interface brief

- Check that all required interfaces are **up/up**.
- Verify R1, R2 LAN and WAN interfaces.

Step 2: Test Local Connectivity

From Branch A:

ping 192.168.10.1

From Branch B:

ping 192.168.20.1

- Confirms PC-to-router connectivity.

Step 3: Test WAN Connectivity

From R1:

ping 10.0.0.2

From R2:

ping 10.0.0.1

- Confirms the WAN link between routers.

Step 4: Check Routing Table

show ip route

- Verify R1 has a route to 192.168.20.0/24.
- Verify R2 has a route to 192.168.10.0/24.

Step 5: Test Branch-to-Branch Connectivity

From Branch A:

ping 192.168.20.10

From Branch B:

ping 192.168.10.10

If ping works but WhatsApp voice does not, continue with UDP/ACL/NAT troubleshooting.

Step 6: Check ACL

show access-lists

show ip interface

Look for rules that block UDP traffic, for example:

deny udp any any

An ACL may allow ICMP/ping while blocking UDP traffic used by voice communication.

Step 7: Check NAT

show ip nat translations

show ip nat statistics

Check whether branch-to-branch traffic is being incorrectly translated.

Step 8: Check UDP/Application Traffic

Verify that UDP traffic between the two branch networks is allowed.

Branch A → R1 → WAN → R2 → Branch B



UDP traffic

Step 9: Correct the Misconfiguration

Modify the ACL to allow the required branch-to-branch traffic while keeping the security policy intact.

Step 10: Verify the Solution

Run:

```
show access-lists  
show ip route  
show ip nat translations  
ping <remote-host>  
tracert <remote-host>
```

Finally, test the **WhatsApp voice call** again and confirm that the call connects successfully.